Matt McManus

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Citizenship: U.S. Citizen | **Location:** New York, NY

Summary

Trader and Head of Prediction Markets at DRW, building AI-driven forecasting and execution systems for event contracts. Leads market selection, quantitative research, automated market making, risk modeling, and production ML. Previously built calibrated LLM forecasting systems at Bridgewater and cross-sectional alphas at Two Sigma.

Education

Massachusetts Institute of Technology — Cambridge, MA
Master of Engineering in Computer Science, GPA: 5.0/5.0 — *Aug. 2023 - Jun. 2024*
- Thesis: Physics-informed neural ODEs for inertial navigation systems with Prof. Alan Edelman (Julia Lab)

Bachelor of Science in Mathematics & Computer Science — *Aug. 2019 - Feb. 2024*
- Graduate/Research: Modeling with ML, LLMs & Beyond, Multi-Agent Comm., Scientific ML, Stat. Learning Theory
- Math/Theory: Probability, Linear Algebra, Optimization, Algorithms, Statistics
- Activities: MIT Varsity Squash, MIT Pokerbots President, MIT Bitcoin Club, HKN Tutor

Experience

DRW - Prediction Markets — New York, NY
Head of Prediction Markets — *Feb. 2026 - Present*
- Lead AI prediction systems from proof of concept through production for event-contract and multi-asset trading.
- Research sports, financial, economic, crypto, and political markets, evaluating venues and regulatory constraints globally.
- Adapt automated market-making and risk models while integrating signals into trading systems and continuous ML pipelines.

Bridgewater Associates - AIA Labs — New York, NY
Machine Learning Engineer — *Sept. 2024 - Dec. 2025*
- Framed LLM reliability as calibration, testing dispersion, selective prediction, and prompt/program search.
- Built time-keyed, no-peek walk-forward evaluation with recency gates and allowlists, assessed Platt/isotonic and LLM-as-Judge methods with Brier and log-loss, and scaled via FastAPI/K8s.

Investment Engineer Intern — *Jun. 2023 - Aug. 2023*
- Built data models, algorithmic systems, and statistical instruments for macroeconomic business-cycle analysis.

Two Sigma — New York, NY
Quantitative Researcher (Part-time) — *Jan. 2024 - Jun. 2024*
- Developed cross-sectional alphas, factor-neutralized by beta/sector/size, and validated via rolling OOS rank IC and IR.
- Designed feature- and learner-level decorrelation (orthogonalization, column subsampling, correlation-penalized loss).
- Built a leakage-safe walk-forward pipeline with rolling normalization, liquidity-weighted scoring, and reproducible backtests/ablations.

MIT CSAIL - Julia Lab (Advisor: Alan Edelman) — Cambridge, MA
Graduate Researcher - Scientific ML & INS — *Sep. 2023 - Jun. 2024*
- Built Julia physics-informed neural ODEs for strapdown INS and cut 3D RMSE by **63%** vs tuned EKF.
- Developed IMU simulation harness enabling **100+** walk-forward tests daily with automated robustness checks.

MIT CSAIL - ALFA (Advisor: Una-May O'Reilly) — Cambridge, MA
Undergraduate Researcher - LLMs for Cyber Defense — *Sep. 2022 - Jun. 2023*
- Built graph-based cyber-defense simulator using GPT-3 for anomaly detection and attack pathing.
- Ran controlled studies across 50+ network topologies and achieved **2x faster** decision latency vs RL baselines.

Selected Research

AIA Forecaster: LLM-Based Judgmental Forecasting — *Bridgewater, 2025*
Technical Report: LLM Agents, Calibration, Forecasting
- LLM forecasting with agentic search, supervisor reconciliation, and calibration matched superforecasters on ForecastBench.
- An ensemble with market consensus beats consensus alone and delivers expert-level AI forecasting at scale.

How Do Transformers "Do" Math? Interpretability for Linear Regression — *MIT, 2024*
Course Research Project / Poster: Mechanistic Interpretability, Probing & Interventions
- Showed transformers encode/use task intermediates (slope w) via features and tied encoding to performance via probing
- Provided causal evidence via reverse probes + interventions (forcing $w \to w'$ predictably shifts outputs)

Full research portfolio: m-mcmanus.com/research/

Technical Skills

ML/AI: PyTorch · TensorFlow · Hugging Face · Transformers/LLMs · Reinforcement Learning · Scientific ML · XGBoost
Programming: Python · Julia · C++ · Scala · Java · Go · SQL/Spark · JavaScript · R
Infrastructure: AWS · Docker · Kubernetes · Git · CI/CD · Distributed Systems · PostgreSQL
Finance: Quantitative Research · Factor Models · Backtesting · Risk Management

Leadership & Achievements

MIT Pokerbots President: Led Harvard-MIT ML poker competition (**250+** students), secured **$100k+** sponsorships
MIT Phi Kappa Theta President: Led fraternity operations and member development
MIT Varsity Squash: Achieved National Team Ranking of **16th** in U.S. (2023-2024), 4-year starter